Assigned Paper: Ng, A. Y., & Jordan, M. I. (2001). On discriminative vs. generative classifiers: A comparison of logistic regression and naive Bayes. In Advances in Neural Information Processing Systems 14 (NIPS 2001), pp. 841–848. MIT Press.

Description: The paper challenges the belief that discriminative classifiers (logistic regression, which models p(y|x)) should always be preferred over generative ones (naive Bayes, which models p(x,y)). It will attempt to prove that logistic regression has the lower asymptotic error but needs on the order of n training examples (n = number of features) to reach it, while naive Bayes reaches its higher asymptotic error after only O(log n) examples. The result is a predicted two-regime learning curve: naive Bayes wins with little data, logistic regression wins with a lot. The authors support this with learning curves on 15 small UCI datasets but never vary n directly, so the log n vs. n rate itself is untested.

Proposed Project Title and Relationship: Does Naive Bayes Really Learn in O(log n)? Measuring the Generative-Discriminative Crossover as Dimensionality Grows. The project replicates the paper's experiment on a text dataset where the feature count n can be set directly, turning the paper's rate claim into a measurable prediction about how the crossover point scales with n.

Problem Statement and Research Questions: While Ng and Jordan (2001) established that naive Bayes converges in O(log n) samples compared to O(n) for logistic regression, their empirical validation lacked confidence bounds and held feature dimensions static, leaving the predicted scaling of the crossover point unverified. We address this gap by evaluating whether naive Bayes statistically outperforms logistic regression at small training sizes m and loses at large m under paired significance testing, whether the crossover threshold m*(n) scales linearly with feature count n as theoretical bounds suggest, and whether the early generative advantage persists under probabilistic log-loss rather than being an artifact of 0-1 misclassification error.

Proposed Dataset and Source: Enron-Spam (Metsis, Androutsopoulos & Paliouras, 2006), taken from the public GitHub mirror (MWiechmann/enron_spam_data) because the original AUEB host's TLS certificate is invalid, contains 33,716 emails labeled ham or spam (17,171 spam) from six Enron folders, and after removing exact duplicates its 30,138 emails allow training sets of up to 22,110 and vocabulary sizes n from 16 to 4,096, so the crossover can be measured at every n. The SMS Spam Collection from the UCI Machine Learning Repository (Almeida, Gómez Hidalgo & Yamakami, 2012; CC BY 4.0), 5,574 English SMS messages (747 spam) or 5,159 after removing duplicates, is kept as a small-data comparison covering n from 16 to 2,048. On both corpora the vocabulary is ranked on a held-out development split so n can be set directly, and binary bag-of-words inputs match the paper's discrete-input setting exactly.

Machine Learning Task and Modeling Approach: We evaluate this dynamic via binary spam classification using paired generative and discriminative linear classifiers trained on binary bag-of-words features across vocabulary sizes n from 16 to 4,096 selected from a held-out development split. Our primary models will directly mirror the paper—Bernoulli naive Bayes with Laplace smoothing (alpha = 1) and unregularized logistic regression with a deterministic policy for linearly separable subsets—complemented by multinomial naive Bayes and L2-regularized logistic regression baselines, all evaluated over 200+ stratified Monte Carlo train/test splits across a log-spaced grid of sample sizes m.

Team Leader and Members: Team leader: Andrew Ngo. Members: Kevin Liu, Bryce Lee.

Initial Responsibilities: Data and experiment pipeline (Andrew Ngo): download and clean the corpus, build vocabularies and feature matrices for each n, write the split/training harness, run all (n, m) experiments and log results. Models and statistical analysis (Kevin Liu): implement the paper's exact naive Bayes and logistic regression variants, define the separable-case policy, run significance tests, estimate m*(n) and fit the scaling models. Theory, writing and presentation (Bryce Lee): map each experiment to the paper's propositions and theorem, produce the figures, draft the paper and build the slides.

Project Roadmap: Week 1 (Oct 20–27): data pipeline, vocabulary splits and paper-faithful models; gate Oct 27: end-to-end run at n = 64. Week 2 (Oct 28–Nov 3): full (n, m) experiment sweep; gate Nov 3: all runs logged. Week 3 (Nov 4–10): significance tests, crossover estimates, scaling fits and figures; gate Nov 10: results frozen. Week 4 (Nov 11–17): paper draft and slides; final Nov 17: paper and presentation submitted.